

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0833

Pineville 345 kV
August 27, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0833. The Interconnection Customer proposes a 160.0 MW solar photovoltaic facility interconnecting at the Pineville 345 kV substation in the Maryland area. The request was received on July 22, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 243 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,773,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Thurmont – Dry Gap Substation 138 kV	W Kelso – Dry Gap 138 kV	135.3	459	8.3
Five Channels Substation – Sheton 138 kV	Five Channels – Edgewood 138 kV	131.9	239	8.9
Metro Center – Skykomish 138 kV L	E Hagerstown – Pineville 138 kV line	130.6	221	5.3
Clark County – Shelton 138 kV	Edgewood – Five Channels Substation 138 kV Line	118.6	246	8.9
Hollister – Metro Ctr 138 kV line	Pineville – E Hagerstown 138 kV	114.1	465	10.6
Pineville Substation 345-138 kV Auto	W Waynesboro 345 kV/138 kV Transformer Bank	110.8	803	32.2
Bon Aqua – Dry Gap Substation 138 kV L	Athalmer Tap – Cameron Substation 138 kV line	103.8	813	11.8
Bon Aqua – Carl Bailey 138 kV line	Cameron – Athalmer Tap 138 kV	103.6	820	11.8
Carl Bailey – Otsego 138 kV	Cameron – Athalmer Tap 138 kV	103.4	831	11.8
Edgewood – Five Channels 138 kV	Five Chnnels – Shelton 138 kV Line	102.5	313	6.9
W Waynesboro 345 kV/138 kV Autotransformer	Pineville 345-138 kV Transformer Bank	101.0	900	28.7
Chambersburg Substation 345 kV/138 kV Transformer Bank	West Waynesboro – E Waynesboro 138 kV	101.0	555	9.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Thurmont – Dry Gap 138 kV L	Rebuild line with higher-capacity conductor	\$5,814,000	9% (\$535,000)
Five Channels Substation – Sheton 138 kV	Rebuild line with higher-capacity conductor	\$1,369,000	13% (\$175,000)
Metro Center – Skykomish 138 kV L	Rebuild line with higher-capacity conductor	\$1,408,000	100% (\$1,408,000)
Clark County – Shelton 138 kV	Rebuild line with higher-capacity conductor	\$1,679,000	13% (\$215,000)
Hollister – Metro Ctr 138 kV	Rebuild line with higher-capacity conductor	\$2,885,000	10% (\$275,000)
Pineville 345-138 kV XFMR	Replace transformer with higher-rated unit	\$12,483,000	15% (\$1,928,000)
Dry Gap – Bon Aqua 138 kV	Reconductor line	\$2,609,000	9% (\$241,000)
Carl Bailey Substation – Bon Aqua 138 kV	Reconductor line	\$2,102,000	9% (\$194,000)
Carl Bailey – Otsego 138 kV	Reconductor line	\$2,559,000	9% (\$237,000)
Edgewood – Five Channels 138 kV	Reconductor line	\$1,831,000	52% (\$945,000)
W Waynesboro 345-138 kV XFMR	Replace transformer with higher-rated unit	\$18,722,000	12% (\$2,240,000)
Chambersburg 345/138kV Auto	Replace transformer with higher-rated unit	\$14,308,000	10% (\$1,380,000)
	Total allocated to project		\$9,773,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.